Data Structures and Algorithms - Assignment 2

1. Write program to implement linear queue in which front and rear starts at 0.

2. Write a program to implement descending stack. (Initialize top = SIZE)

3. Implement circular queue using counter method (maintain count of elements in a queue and use it for checking empty and full conditions)

4. Implement following algorithms for multi digit operands (use space to seperate operands)

- postfix evaluation
- prefix evaluation

Optional

5. Paperwork: Convert following examples from Infix to Prefix and Postfix.

- K + L - M * N + (O $ P) * W / U /
- V * T + Q (A + B) * C - (D - E) * (F + G)

6. Paperwork: Convert following expression to Postfix and Prefix using stack.

- 5+9-4*(8-6/2)+1$(7-3)

7. https://leetcode.com/problems/valid-parentheses

SUNBEAM